PCBC H Parametric & Sequences Test (Practice C) **Name**_____ _____/16

SHOW ALL WORK FOR FULL CREDIT. ***Non-Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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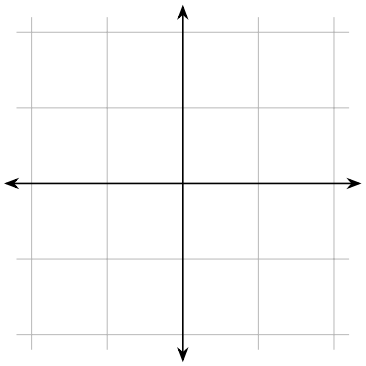
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Signature

Write the pair of parametric equations in rectangular form, including the domain. Then sketch the curve, including direction. (3 points each)

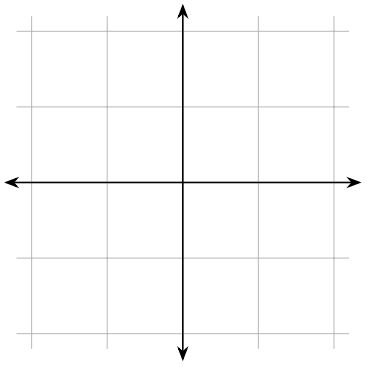
1. $x = t + 2, \quad y = t^2 + 4t + 1, \quad -4 \le t \le 0$

SIMPLIFIED rectangular form: _____



2. $x = t^2, \quad y = t - 1, \quad -2 \le t \le 2$

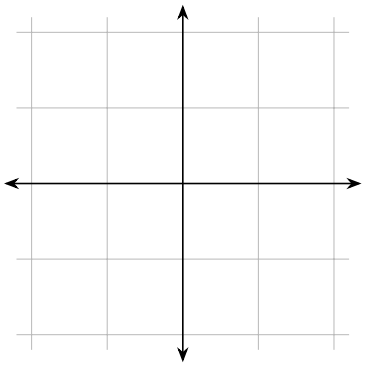
rectangular form: _____



Write the pair of parametric equations in rectangular form. Then sketch the curve. (3 points each)

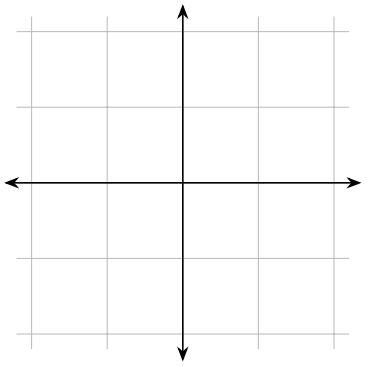
3. $x = 3\sin t, \quad y = 3\cos t, \quad 0 \le t \le 2\pi$

rectangular form: _____



4. $x = 2\sec t, \quad y = 3\tan t, \quad t \text{ in } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

rectangular form: _____



Convert the rectangular equation to parametric form. Give two different sets of answers. (2 points)

5. $y = x^2 - 8x + 10$

Write down the first five terms of the sequence. (2 points)

6. $\{a_n\} = \left\{(-1)^n \frac{n+2}{2^n}\right\}$

Extra practice: the other sequence questions #6 could be

Not part of the 16 points. One of these types is the one sequence question; worked in the key.

A sequence is defined recursively. Write the first five terms.

E1. $a_1 = 3; \quad a_n = a_{n-1} + n$

Write out the sum.

E2. $\sum_{k=1}^{n} (-1)^k k$

Express the sum using summation notation.

E3. $1 - \frac{1}{8} + \frac{1}{64} - \frac{1}{512} + \cdots + (-1)^8 \left(\frac{1}{8}\right)^8$

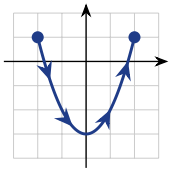
Answer Key

Work on the left, the reason beside it. Boxed = final answer.

Day 1: Non-Calculator

1. $x = t+2,\ y = t^2+4t+1,\ -4 \le t \le 0$

t	(x,y)
-4	$(-2,1)$
-3	$(-1,-2)$
-2	$(0,-3)$
-1	$(1,-2)$
0	$(2,1)$



left to right; solid dots at both ends

$t = x-2$ — *one answer for t*

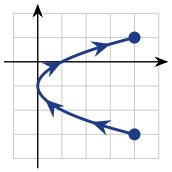
$y = (x-2)^2+4(x-2)+1$ — *substitute*

$y = x^2-4x+4+4x-8+1$ — *SIMPLIFIED: y alone, simplify*

$\boxed{y = x^2-3 \text{ for } x \text{ in } [-2,2]}$ — *domain from the x column*

2. $x = t^2,\ y = t-1,\ -2 \le t \le 2$

t	(x,y)
-2	$(4,-3)$
-1	$(1,-2)$
0	$(0,-1)$
1	$(1,0)$
2	$(4,1)$



bottom to top: from $(4,-3)$ to $(4,1)$

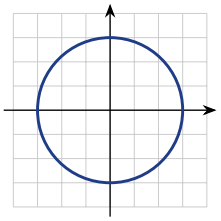
$t = y+1$ — *isolate t in y (solving $x = t^2$ gives $\pm$)*

$\boxed{x = (y+1)^2 \text{ for } x \text{ in } [0,4]}$ — *sideways parabola; x column min 0 (turning point), max 4*

y alone would need $\pm$, so leave it as $x = (y+1)^2$. Watch the turning point: x goes down to 0, not just the ends.

3. $x = 3\sin t,\ y = 3\cos t,\ 0 \le t \le 2\pi$

t	(x,y)
0	$(0,3)$
$\frac{\pi}{2}$	$(3,0)$
π	$(0,-3)$
$\frac{3\pi}{2}$	$(-3,0)$
2π	$(0,3)$



circle, radius 3

$\sin t = \frac{x}{3}, \quad \cos t = \frac{y}{3}$

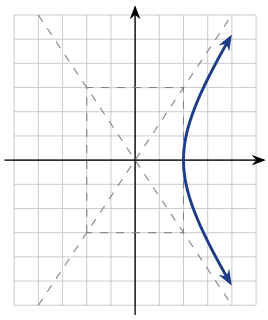
$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$ $\qquad \sin^2 t + \cos^2 t = 1$

$\boxed{x^2 + y^2 = 9}$ $\qquad$ *whole circle: done*

4. $x = 2\sec t,\ y = 3\tan t,\ t$ in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

t	(x,y)
$-\frac{\pi}{4}$	$(2\sqrt{2},-3)$
0	$(2,0)$
$\frac{\pi}{4}$	$(2\sqrt{2},3)$

all x positive: right branch only



right branch only; end arrowheads: it keeps going

$\sec t = \frac{x}{2}, \quad \tan t = \frac{y}{3}$

$\frac{x^2}{4} - \frac{y^2}{9} = 1$ $\qquad 1 + \tan^2 t = \sec^2 t$

$\boxed{\frac{x^2}{4} - \frac{y^2}{9} = 1 \text{ for } x \text{ in } [2,\infty)}$ $\qquad$ *left-right hyperbola, one branch: the domain cuts it*

Never "for y in". No need to solve for y here.

5. $y = x^2 - 8x + 10$

$\boxed{x = t,\ y = t^2 - 8t + 10}$

$x = t + 4:\ y = (t+4)^2 - 8(t+4) + 10$ $\qquad$ *pick a shift, substitute*

$y = t^2 + 8t + 16 - 8t - 32 + 10$ $\qquad$ *simplify*

$\boxed{x = t + 4,\ y = t^2 - 6}$

Both for t in $(-\infty,\infty)$ (assumed). Give both $x =$ and $y =$.

6.

$(-1)^1\frac{3}{2},\ (-1)^2\frac{4}{4},\ (-1)^3\frac{5}{8},$ $\qquad$ *plug in $n = 1$ to 5*

$(-1)^4\frac{6}{16},\ (-1)^5\frac{7}{32}$

$\boxed{-\frac{3}{2},\ 1,\ -\frac{5}{8},\ \frac{3}{8},\ -\frac{7}{32}}$ $\qquad$ *reduce: $\frac{4}{4} = 1$, $\frac{6}{16} = \frac{3}{8}$; odd n negative*

Extra practice (not scored)

E1. $a_1 = 3,\ a_n = a_{n-1} + n$

$a_2 = 3 + 2 = 5,\ \ a_3 = 5 + 3 = 8$ $\qquad$ *add the term number n to the previous term*

$a_4 = 8 + 4 = 12,\ \ a_5 = 12 + 5 = 17$

$\boxed{3,\ 5,\ 8,\ 12,\ 17}$

E2. $\sum_{k=1}^{n} (-1)^k k$

$(-1)^1(1)+(-1)^2(2)+(-1)^3(3)+\cdots+(-1)^n n$ — *$k=1$ is negative*

$\boxed{-1+2-3+\cdots+(-1)^n n}$ — *don't add*

E3. $1-\frac{1}{8}+\frac{1}{64}-\cdots+(-1)^8\left(\frac{1}{8}\right)^8$

$1=\left(\frac{1}{8}\right)^0,\ -\frac{1}{8}=(-1)^1\left(\frac{1}{8}\right)^1$ — *the first term is power 0, so start at $k=0$*

$\boxed{\sum_{k=0}^{8}(-1)^k\left(\frac{1}{8}\right)^k}$ — *the last term gives the upper index 8*